

An Exact 10^{10} -Term Boundary for Jumps 14 and 15 and the First Jump 16 in OEIS A260194

Xiaojian Zeng

PBC School of Finance, Tsinghua University, Beijing 100190, China
xiaojianzeng@foxmail.com

September 6, 2026

Abstract

Let $a_1 = a_2 = a_3 = 1$ and $a_{n+1} = a_n + \gcd(a_n, a_{n-2})$ for $n \geq 3$, and put $J_n = a_{n+1} - a_n$. Exact exhaustive computation shows that $J_n \notin \{14, 15\}$ for every $1 \leq n \leq 9,999,999,999$. It also locates the first occurrence of jump 16 at $n = 399,815,271$. We state the finite result precisely, derive an equivalent recurrence used for an independent check, and provide constant-memory implementations and endpoint certificates. The assertion that every positive integer occurs as a jump remains unresolved; in particular, this computation does not exclude 14 or 15 after the stated boundary.

1 Statement and finite scope

Consider the unique integer sequence

$$a_1 = a_2 = a_3 = 1, \quad a_{n+1} = a_n + \gcd(a_n, a_{n-2}) \quad (n \geq 3), \quad (1)$$

and define $J_n = a_{n+1} - a_n$ for $n \geq 1$. Thus $J_1 = J_2 = 0$ and $J_n = \gcd(a_n, a_{n-2})$ for $n \geq 3$. The OEIS entry A260194 asks whether every positive integer occurs among the J_n [1]. It reports that, in a computation of the first 100,000 terms, the maximum jump is 21 and the values 14, 15, and 16 are missing among $1, \dots, 21$.

Theorem 1 (Exact finite boundary). *For the orbit defined by (1),*

1. $J_n \notin \{14, 15\}$ for every $1 \leq n \leq 9,999,999,999$;

2.

$$\min\{n \geq 1 : J_n = 16\} = 399,815,271.$$

At this index,

$$(a_{n-2}, a_n, a_{n+1}) = (920,798,928, 920,798,992, 920,799,008).$$

The theorem is a statement about a finite prefix of one deterministic orbit. It does not assert that 14 or 15 never occurs, and it neither proves nor disproves the universal-coverage question in A260194.

The explicitly reported computational horizon in A260194 is 10^5 terms; Theorem 1 treats 10^{10} terms, a factor of 10^5 farther, and removes 16 from the list of values not yet observed. Targeted searches on September 6, 2026 for the A-number, the exact recurrence, the OEIS question, and the witness index found no equivalent larger published computation beyond the OEIS entry. This dated, limited search is not an exhaustive literature review and is not a claim of priority.

2 Identity and exhaustive computation

For every $n \geq 3$, telescoping two adjacent jumps gives

$$a_n - a_{n-2} = J_{n-2} + J_{n-1}. \quad (2)$$

The Euclidean algorithm therefore yields the equivalent update

$$J_n = \gcd(a_n, a_{n-2}) = \gcd(a_{n-2}, J_{n-2} + J_{n-1}), \quad a_{n+1} = a_n + J_n. \quad (3)$$

This form provides a check that is structurally different from computing the greatest common divisor of the two large orbit terms directly.

Set $T = 10^{10}$. The full computation initializes

$$(a_{n-2}, a_{n-1}, a_n) = (1, 1, 1), \quad (J_{n-2}, J_{n-1}) = (0, 0)$$

at $n = 3$. For each integer $n = 3, \dots, T - 1$, it performs the following exact operations:

1. form $s = J_{n-2} + J_{n-1}$ and check $a_n - a_{n-2} = s$;
2. compute $J_n = \gcd(a_{n-2}, s)$, then set $a_{n+1} = a_n + J_n$;
3. update the counts for 14, 15, and 16, retain the first witness for 16, and shift the rolling state by one index.

Together with the two initial zero jumps, this loop covers exactly $J_1, \dots, J_{T-1}$. Induction using (3) shows that the rolling state at every iteration is the state of (1); hence zero target counts are exhaustive finite nonoccurrence certificates, rather than samples.

The transformed-identity run returned

$$\#\{n < T : J_n = 14\} = 0, \quad \#\{n < T : J_n = 15\} = 0, \quad \#\{n < T : J_n = 16\} = 28, \quad (4)$$

with first jump 16 at the index in Theorem 1. Its largest jump on the prefix was 64, and its final three terms were

$$(a_{T-2}, a_{T-1}, a_T) = (23,030,608,785, 23,030,608,788, 23,030,608,792).$$

All additions in the C++ implementations were checked before execution; the final value is far below $2^{64} - 1$. There was no floating-point arithmetic or randomized step.

The local witness also satisfies

$$\begin{aligned} 920,798,992 - 920,798,928 &= 64, \\ \gcd(920,798,992, 920,798,928) &= 16, \\ 920,799,008 - 920,798,992 &= 16. \end{aligned}$$

These equalities verify the local jump once the three integers are known to be orbit terms. The exhaustive recurrence computation supplies that latter link, as well as the absence of an earlier jump 16.

3 Independent implementations and reproducibility

The accompanying source and machine-readable output provide three checks with different arithmetic or bookkeeping:

Exact Python prefix. `stream_orbit.py` uses arbitrary-precision Python integers, `math.gcd`, a rolling triple, and assertions for both (1) and (3). With term count 399,815,272, it exhausts all jumps through the first 16 and writes `stream_orbit_399815272.json`.

Full histogram scan. `stream_orbit_independent.cpp` uses checked `uint64_t`, `std::gcd`, a rolling triple, and a complete jump histogram. With term count 10^{10} , its output is `stream_orbit_independent_10000000000.json`.

Minimal target scan. `verify_boundary_independent.cpp` has no histogram or shared diagnostic machinery. It uses a handwritten Euclidean algorithm and retains only the rolling triple, the maximum jump, target counts, and first witnesses. Its 10^{10} -term output is `verify_boundary_independent_10000000000.json`.

The two full C++ outputs agree on the final triple, maximum jump, target counts, and first-16 certificate. The exact Python prefix independently agrees on the least-index certificate. The C++ sources can be compiled with, for example, `c++ -O3 -DNDEBUG -std=c++17`; each executable takes the desired term count as its sole argument. The archived verification environment used Apple clang 16.0.0 on 64-bit ARM Darwin. The programs abort before a possible integer overflow.

4 Limitations and assistance disclosure

The zero counts in (4) end at $J_{9,999,999,999}$. Either 14 or 15 may occur later. More generally, a finite orbit computation cannot establish that every positive jump eventually occurs. The computation also supplies no invariant that permanently excludes a target value. Finally, the literature comparison above records only the specified searches on one date; it does not establish originality.

An OpenAI Codex language model assisted with implementation design, cross-check organization, evidence synthesis, and drafting and editing this note. Language-model output was not treated as mathematical proof: the claim is restricted to the explicitly enumerated exact computation, whose source, loop endpoints, output fields, and limitations are exposed here. The project also supplied Zijian Zeng’s manuscript on AI chatbot code generation [2]. It is cited only in connection with this assistance disclosure, not as evidence for the theorem, prior art, open status, or originality. No abstract, venue, or DOI metadata for that item was supplied.

References

- [1] F. T. Adams-Watters. A260194. The On-Line Encyclopedia of Integer Sequences, 2015. Entry last modified September 5, 2026; accessed September 6, 2026.
- [2] Z. Zeng. AI Chatbot for Code Generation. User-supplied unpublished manuscript; no abstract, venue, or DOI metadata supplied, 2025.